Electric circuit

Kirchhoff’s laws, energy of electromagnetic field, AC, LC circuit

Kirchhoff's rules (基尔霍夫定律)

- Make analysis of complicated circuits.
- Kirchhoff's first rule (junction rule, 节点定律): the algebraic sum of the current is zero.

$$\frac{\mathrm{d}q}{\mathrm{d}t} = \sum I_{\mathrm{in}} - \sum I_{\mathrm{out}} = \sum_i I_i = 0$$

- Kirchhoff's second law (loop rule, 回路定律): for an arbitrary closed path, the algebraic sum of the potential changes through the whole loop is zero.

$$-\oint \mathbf{E} \cdot \mathrm{d}\mathbf{l} = \sum V - \sum (\mathcal{E} \boxed{- Ir}) = 0$$

Kirchhoff (1824-1887)

Example: Wheatstone bridge (惠斯通电桥)

- For junction b and d:

$$I_x = I_1 + I_g, I_S = I_2 - I_g$$

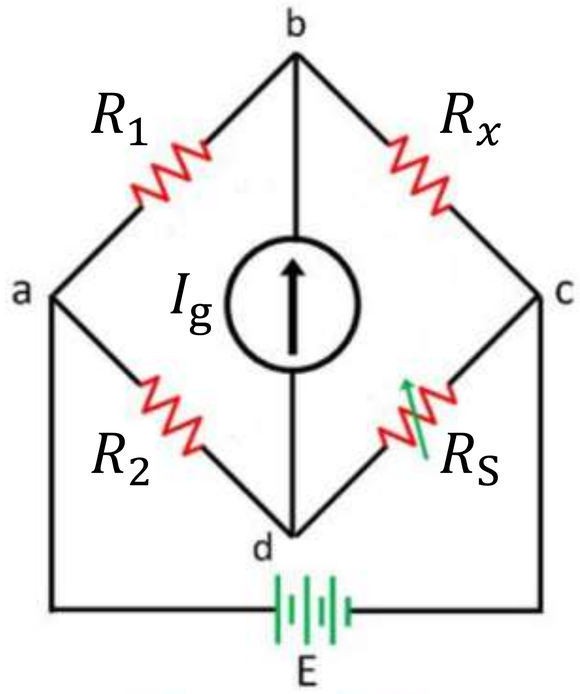
- For loop abd and bcd:

$$I_1 R_1 + I_g R_g = I_2 R_2, I_x R_x = I_S R_S + I_g R_g$$

- When $I_g = 0$

$$\frac{R_1}{R_x} = \frac{R_2}{R_S} \quad \rightarrow \quad R_x = \frac{R_1}{R_2} R_S$$

- If $I_g \neq 0$, R_x can also be obtained with I_g.
- Wheatstone bridge is sensitive and accurate: superior to voltammetry (伏安法)

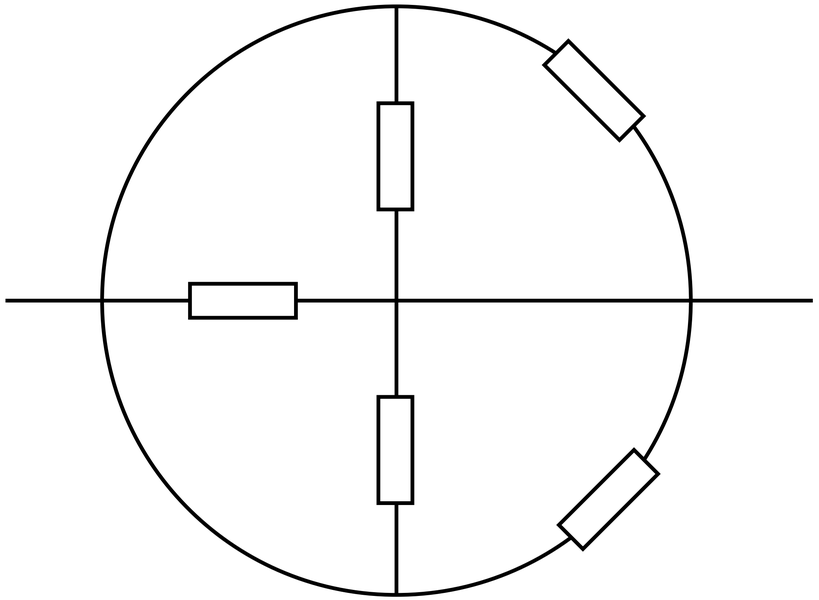


Wheatstone Bridge

填空题 1分

设置

As is shown, the resistance of each resistor is R. Determine the equivalent resistance between A and B: [填空1] R.



作答

*Analysis of complex circuit

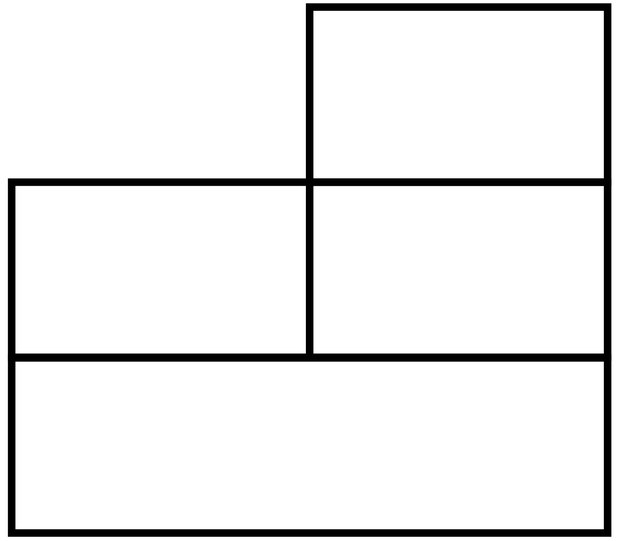
- Dimension of a network
  - A branch: $N = 2$, V and I

$$N = 2b$$

- Constraints
  - A junction: $l = 1$; a loop: $l = 1$
  - ※ Every loop must have at least one exclusive (独占) branch.
  - A determinated branch: $l = 1$
- Degree of freedom of a circuit network

$$s = 2b - (j + k + m)$$

- If $s = 0$, the circuit can be completely solved.



Electric work and power

- For an infinitesimal charge $\mathrm{d}q$ flowing through a voltage V, the work done on the charge:

$$\mathrm{d}W = V\mathrm{d}q$$

- The electric power can be derived:

$$P = \frac{\mathrm{d}W}{\mathrm{d}t} = \frac{V\mathrm{d}q}{\mathrm{d}t} = VI$$

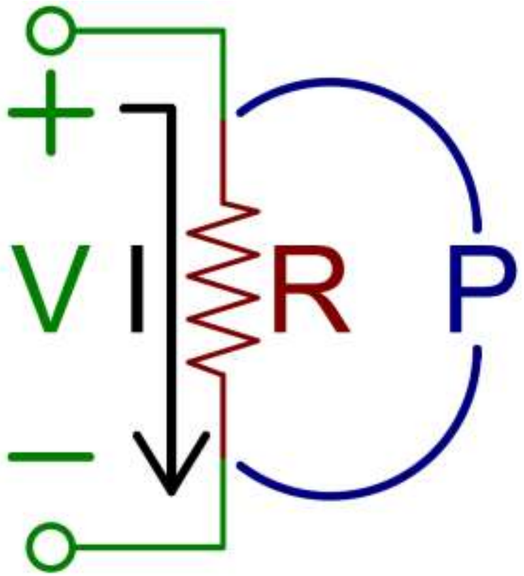
  - This is electric energy consumption

- For a power source:

$$\mathrm{d}W = \mathcal{E}\mathrm{d}q$$

$$P = \mathcal{E}I$$

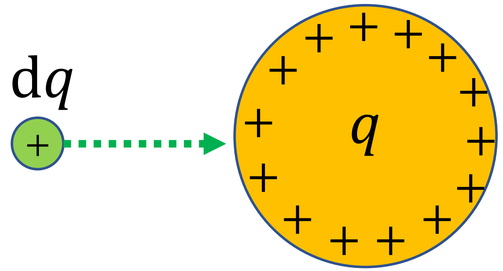
  - This is electric energy generation



Energy of electromagnetic field

- Charge of a conductor

$$\mathrm{d}W = v\mathrm{d}q = Cv\mathrm{d}v \;\rightarrow\; \underline{U = C\int_0^V v\mathrm{d}v = \frac{1}{2}CV^2}$$



- Electric energy in a plate capacitor:

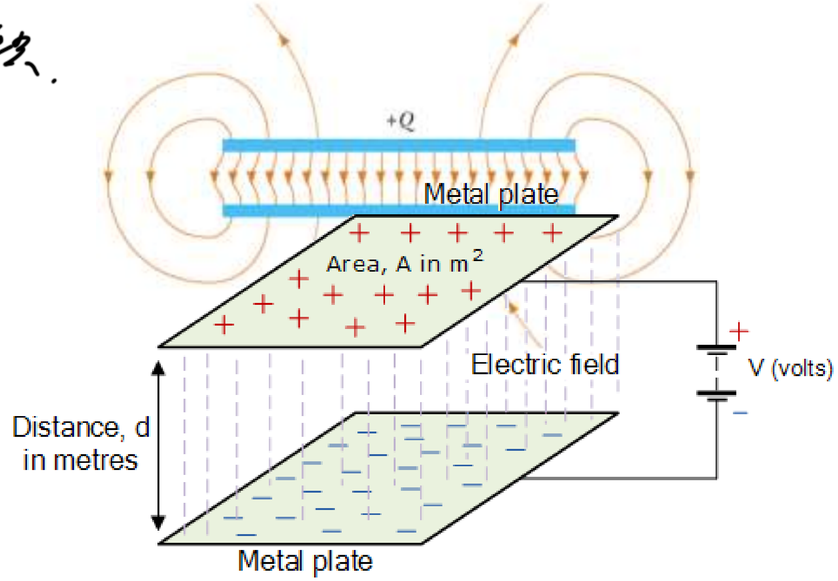
$$U = \frac{1}{2}CV^2 = \frac{\varepsilon A}{2d}(Ed)^2 = \underline{\frac{1}{2}\varepsilon E^2 Ad}$$

体积.

- Ad is the volume between the plates:

$$u = \frac{U}{Ad} = \frac{1}{2}\varepsilon E^2 = \frac{1}{2}\mathbf{D}\cdot\mathbf{E}$$

- Energy density (能量密度): a local quantity



Energy storage of inductor

- *Permittivity tensor (介电张量)

$$\mathbf{D} = \boldsymbol{\varepsilon}\mathbf{E}, u = \frac{1}{2}\mathbf{E}^{\mathrm{T}}\boldsymbol{\varepsilon}\mathbf{E}$$

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_{xx} & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_{yy} & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_{zz} \end{pmatrix}$$

- For a circuit carrying a varying current

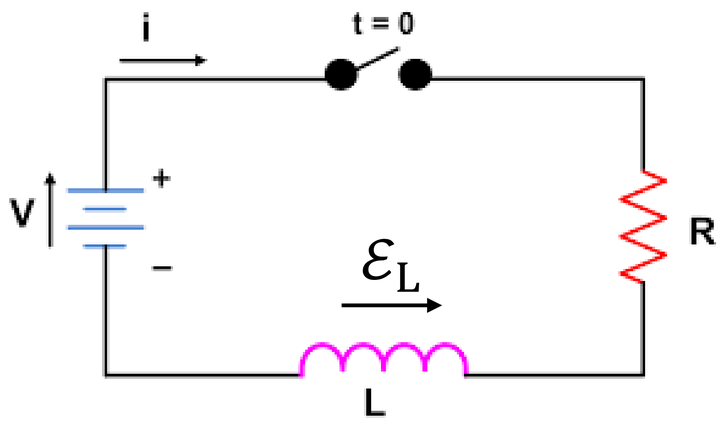
$$V_{\mathrm{R}} = IR = V + \mathcal{E}_{\mathrm{L}}$$

$$P = \underline{VI} = \underline{I^2R - \mathcal{E}_{\mathrm{L}}I} = P_{\mathrm{R}} + P_{\mathrm{L}}$$

- Energy stored in the inductor: induced EMF

$$\mathcal{E}_{\mathrm{L}} = -L\frac{\mathrm{d}I}{\mathrm{d}t} \rightarrow P_{\mathrm{L}} = -\mathcal{E}_{\mathrm{L}}I = LI\frac{\mathrm{d}I}{\mathrm{d}t}$$

$$\mathrm{d}U = P\mathrm{d}t = LI\mathrm{d}I \rightarrow \underline{U = \int_0^I LI\mathrm{d}I = \frac{1}{2}LI^2}$$

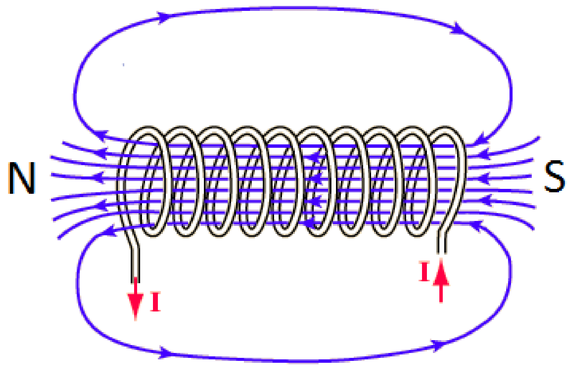


Energy storage of magnetic field

- Energy in capacitor is stored in electric field, energy in inductor is stored in magnetic field.
- Uniform magnetic field by straight solenoid:

$$U = \frac{1}{2}LI^2 = \frac{1}{2}\frac{\mu N^2 A}{l}\left(\frac{Bl}{\mu N}\right)^2 = \frac{B^2 Al}{2\mu}$$

$$u = \frac{U}{Al} = \frac{B^2}{2\mu} = \frac{1}{2}\mathbf{B}\cdot\mathbf{H}$$



- *Energy storage due to mutual inductance:

$$P_1 \mathrm{d}t = MI_1 \mathrm{d}I_2, P_2 \mathrm{d}t = MI_2 \mathrm{d}I_1$$

$$\mathrm{d}U = M(I_1 \mathrm{d}I_2 + I_2 \mathrm{d}I_1) \rightarrow U = MI_1 I_2$$

[Example] A long straight cylindrical wire of radius R and permeability μ carries current of I. Determine the energy of magnetic field inside the wire per unit length.

[Solution] Select a circular loop perpendicular to the axis with its center on the axis:

$$\oint \mathbf{H}\cdot \mathrm{d}\mathbf{l} = 2\pi r H = I_{\text{in}} \quad \rightarrow \quad B = \mu H = \frac{\mu}{2\pi r}\frac{\pi r^2}{\pi R^2} I$$

The magnetic energy density is

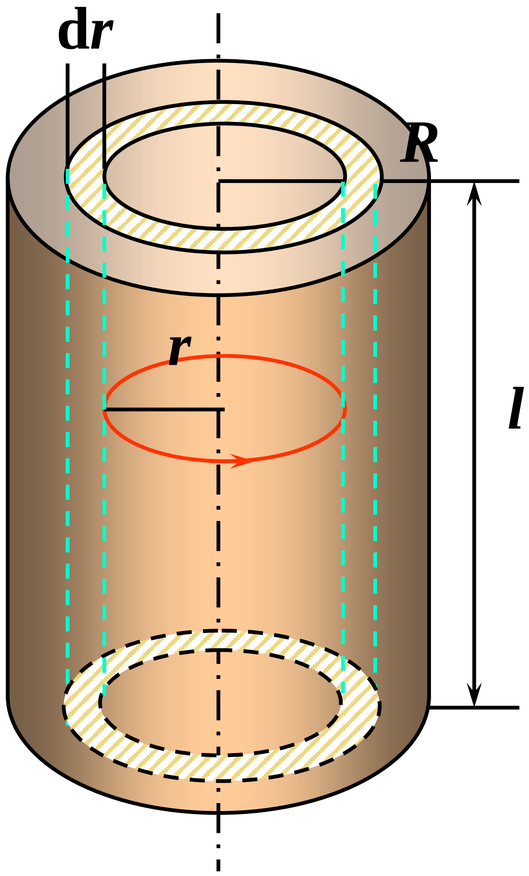
$$u = \frac{B^2}{2\mu} = \frac{\mu I^2 r^2}{8\pi^2 R^4}$$

$u \cdot \pi R^2 l$

Divide the wire into coaxial tubes of thickness $\mathrm{d}r$:

$$\mathrm{d}V = 2\pi r l \mathrm{d}r \quad \rightarrow \quad \mathrm{d}U = u\mathrm{d}V = \frac{\mu I^2 l}{4\pi R^4} r^3 \mathrm{d}r$$

$$U = \int_0^R \frac{\mu I^2 l}{4\pi R^4} r^3 \mathrm{d}r = \frac{\mu I^2}{16\pi} l \quad \rightarrow \quad \frac{U}{l} = \frac{\mu I^2}{16\pi}$$

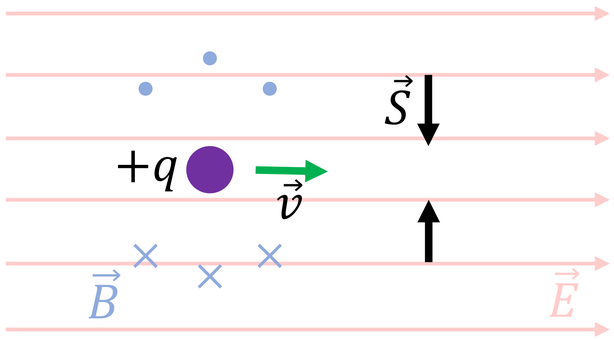


[Discussion] If we try to calculate the energy of the whole space, we will find it approaching infinity. This is because we assume the wire is infinitely long, which is impossible.

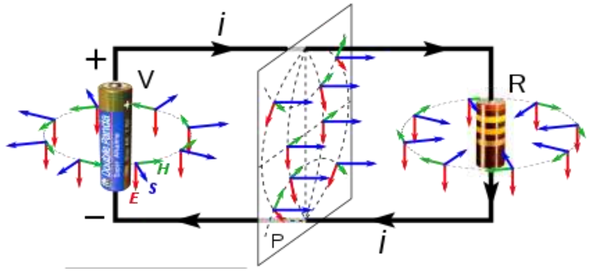
Electromagnetic energy transmission

- Coexisting electric and magnetic field make up energy transmission: Poynting (坡印廷) vector

$$\mathbf{S} = u\mathbf{v} = \mathbf{E} \times \mathbf{H}$$



- Unit: $\mathrm{W/m^2}$, “power density” (功率密度)
- *How does a charge gain energy in electric field?
- *Energy transmission in circuits
  - Energy is transmitted not in the wire but along the electromagnetic field outside the wire.
  - The accumulation of energy of electric field is done by Poynting vector pointing inwards.



*LC circuit, electromagnetic oscillation

- LC circuit

$$V_{\mathrm{C}} + V_{\mathrm{L}} = \frac{Q}{C} - \mathcal{E}_{\mathrm{L}} = \frac{Q}{C} + L\frac{\mathrm{d}I}{\mathrm{d}t} = 0$$

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \frac{\mathrm{d}^2Q}{\mathrm{d}t^2} \rightarrow \frac{\mathrm{d}^2Q}{\mathrm{d}t^2} + \frac{1}{LC}Q = 0$$

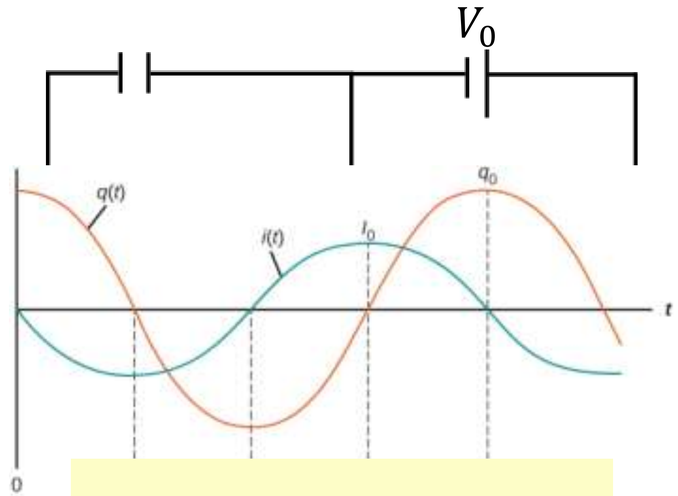
- This is a kind of "SHM".

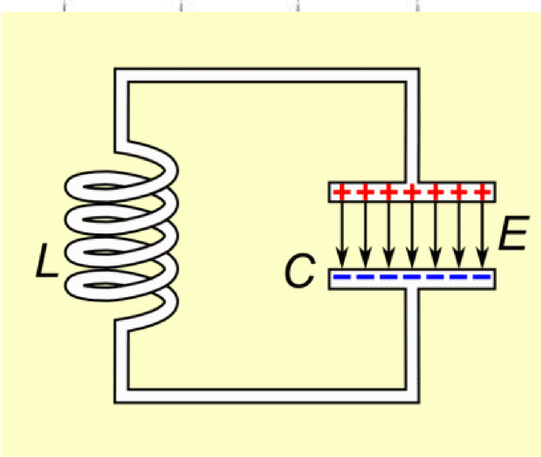
$$Q = Q_0\cos(\omega t + \varphi) \ldots\ldots \omega = \sqrt{1/LC}$$

- Energy of LC oscillation

$$\begin{cases} U_{\mathrm{E}} = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{Q_0^2}{2C}\cos^2(\omega t + \varphi) \\ U_{\mathrm{B}} = \frac{1}{2}LI^2 = \frac{1}{2}LI_0^2\sin^2(\omega t + \varphi) \end{cases}$$

$$U_{\mathrm{E}} + U_{\mathrm{B}} = U$$





*Alternating current (AC, 交流电)

- Sinusoidal alternating current (正弦交流电):

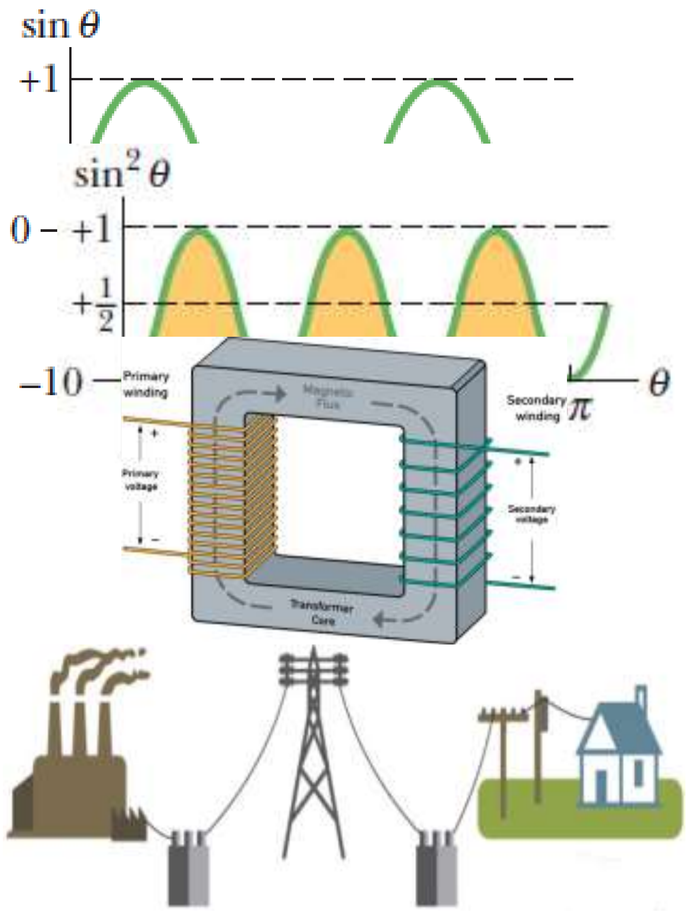
$$V = V_0 \cos \omega t \,, I = I_0 \cos \omega t \ldots \omega = 2\pi f$$

- Average power of AC:

$$\bar{P} = \frac{1}{T}\int_0^T VI\mathrm{d}t = \frac{1}{2}V_0 I_0$$

- Transformer (变压器)

$$\frac{\mathrm{d}\Phi_\mathrm{M}}{\mathrm{d}t} = \frac{\mathcal{E}_\mathrm{P}}{N_\mathrm{P}} = \frac{\mathcal{E}_\mathrm{S}}{N_\mathrm{S}} \rightarrow \frac{V_\mathrm{S}}{V_\mathrm{P}} = \frac{N_\mathrm{S}}{N_\mathrm{P}}$$

- Transmission of electric energy:

$$P_\mathrm{P} = P_\mathrm{S} \rightarrow V_\mathrm{P} I_\mathrm{P} = V_\mathrm{S} I_\mathrm{S} \rightarrow \frac{I_\mathrm{S}}{I_\mathrm{P}} = \frac{N_\mathrm{P}}{N_\mathrm{S}}$$